

# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE

### Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and maintained **Java Spring Boot REST APIs** for enterprise systems, processing approximately **1,500 requests per minute** during peak hours.
- Implemented and debugged **Hibernate** entities and JPQL queries to support complex data retrieval and persistence logic across relational databases.
- Authored **JUnit and Mockito tests** for Java components, achieving over **85% code coverage** for critical business logic modules.
- Contributed to the design and implementation of sustainable engineering solutions, ensuring compliance with **Eurocodes and Irish Building Regulations**.
- Produced detailed **design calculations, specifications, and construction drawings** for structural elements, adhering to industry standards.
- Participated in site inspections, documenting observations and contributing to comprehensive **inspection reports** for ongoing projects.
- Containerized Java applications using **Docker**, facilitating consistent deployment across development and staging environments on **AWS EC2**.

## SKILLS

**Engineering Standards** – Eurocodes, Irish Building Regulations, Design Standards

**CAD & BIM Software** – Revit, AutoCAD, Civil 3D, Site 3D, BIM, Bluebeam

**Programming Languages** – Java, Python, SQL

**Frameworks & Libraries** – Spring Boot, Hibernate, JPA, pandas, NumPy

**Cloud & DevOps** – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

**Databases** – PostgreSQL, MySQL, Oracle

**Tools & Methodologies** – Jira, Confluence, Agile, Scrum, Design Calculations

## PROJECTS

### Structural Design Calculation Tool (HCL Technologies)

- Developed a **Java Spring Boot** application to automate structural load calculations, reducing manual computation time by **40%**.
- Integrated **Hibernate and PostgreSQL** to manage a library of material properties and design codes, ensuring data integrity for calculations.
- Designed user interfaces for inputting project parameters and visualizing calculation results, improving usability for engineering teams.

### Construction Drawing Automation Script (HCL Technologies)

- Authored **Python scripts** to automate the generation of standard construction drawings using **AutoCAD** APIs, increasing drawing output speed by **30%**.
- Developed data parsing modules to extract information from specification documents, ensuring accurate data mapping to drawing elements.
- Collaborated with senior engineers to define drawing templates and standards, ensuring consistency across generated outputs.

## EDUCATION

### M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

### B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

## CERTIFICATIONS

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- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)